

MENGJIN, China

WMO: 570710

Lat: 34.8018N

Lon: 112.4676E

Elev: 1082

StdP: 14.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	23.1	-9.8	3.4	33.5	-5.0	4.4	33.7	23.1	39.9	20.9	40.2	5.2	40	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	95.2	71.8	92.6	72.2	90.0	72.2	80.4	88.0	79.0	86.1	77.6	84.4	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	154.2	85.1	77.1	146.8	83.4	75.7	140.1	82.3	45.3	88.0	43.7	86.4	42.2	84.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	16.0	13.6	DB	16.6	100.7	3.5	2.4	14.0	102.4	12.0	103.8	10.0	105.1	7.4	106.9	
			WB	13.1	82.7	3.2	1.4	10.8	83.7	9.0	84.6	7.2	85.3	4.8	86.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	33.8	39.4	50.1	61.2	70.8	78.6	80.6	78.2	70.3	60.9	48.3	37.7
	DBStd	17.46	6.14	7.18	8.06	7.48	6.09	5.50	4.63	5.28	5.77	6.31	7.65	6.23
	HDD50	1417	502	302	99	9	0	0	0	0	4	118	383	
	HDD65	3847	967	717	465	155	19	1	0	1	156	501	847	
	CDD50	4797	0	5	102	346	643	858	950	873	609	342	68	1
	CDD65	1751	0	0	3	42	197	409	485	409	177	29	0	0
	CDH74	16134	0	0	41	365	1634	4242	4981	3658	1053	157	3	0
	CDH80	6226	0	0	6	95	579	1892	2055	1306	272	21	0	0
Wind	WSAvg	6.3	6.3	6.8	7.3	7.2	6.9	6.1	5.7	5.5	5.3	5.6	6.1	6.4
Precipitation	PrecAvg	23.7	0.3	0.6	1.0	1.6	2.1	2.3	5.7	3.4	3.7	1.9	0.9	0.3
	PrecMax	41.3	1.8	2.5	2.4	4.4	7.4	6.2	10.4	7.5	9.0	5.2	3.5	1.4
	PrecMin	14.0	0.0	0.0	0.0	0.2	0.1	0.1	1.0	0.2	0.6	0.1	0.0	0.0
	PrecStd	5.6	0.4	0.6	0.7	1.1	1.8	1.7	2.7	1.8	2.2	1.4	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	63.7	78.8	88.0	95.5	99.0	97.5	96.0	91.7	82.8	73.1	59.3
		MCWB	41.0	47.5	55.0	64.3	63.7	68.9	75.5	76.0	70.4	60.1	53.9	43.3
	2%	DB	51.3	58.8	72.2	82.0	89.9	95.5	94.4	92.0	86.0	78.2	67.7	54.5
		MCWB	37.9	44.5	53.0	59.5	63.6	68.7	75.5	76.6	69.7	59.8	51.5	40.2
	5%	DB	47.2	54.8	67.4	78.0	86.2	92.6	91.8	89.1	82.8	74.6	63.4	50.8
		MCWB	35.5	41.9	50.9	59.3	63.3	68.5	76.0	76.1	67.7	58.8	49.7	38.4
	10%	DB	43.6	51.0	63.1	74.0	82.6	89.6	89.2	86.6	79.8	71.2	59.6	47.4
		MCWB	33.6	40.3	48.9	57.1	62.9	68.4	76.2	75.4	66.9	57.4	47.8	36.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	49.8	60.2	68.6	72.1	77.4	82.7	82.1	77.3	65.6	57.2	45.7
		MCDB	54.4	57.9	69.6	76.7	80.8	87.4	89.4	89.4	83.5	73.8	67.5	54.4
	2%	WB	38.6	46.0	55.6	65.2	69.6	75.4	81.0	80.3	73.9	63.5	53.9	42.2
		MCDB	49.0	54.9	69.5	76.6	79.7	86.1	88.5	87.6	80.9	71.9	62.9	50.6
	5%	WB	36.4	43.6	52.7	62.3	67.8	73.7	79.5	79.0	71.6	61.9	51.7	40.1
		MCDB	45.1	52.1	64.5	72.5	78.5	84.6	87.1	85.4	77.9	69.3	60.0	48.3
	10%	WB	34.5	41.3	50.1	59.9	66.1	72.1	78.3	77.6	69.5	60.3	49.5	38.0
		MCDB	42.0	49.6	61.1	70.0	76.7	82.8	85.6	83.6	76.4	67.6	57.1	45.7
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.1	16.0	17.5	17.7	16.9	13.3	12.5	13.3	13.9	13.8	13.5
		MCDBR	19.7	21.0	23.0	24.1	24.5	22.3	17.9	16.3	18.5	19.8	20.4	19.9
		MCWBR	11.4	11.7	11.3	10.4	8.9	7.1	6.0	5.7	6.7	8.1	9.9	11.2
	5% WB	MCDBR	18.2	18.5	20.8	19.4	17.4	16.5	14.1	13.5	14.9	14.9	17.4	18.0
	MCWBR	10.8	10.9	11.1	10.4	8.4	6.8	6.1	5.9	6.9	7.2	9.2	10.8	
Clear-Sky Solar Irradiance	taub	0.548	0.629	0.631	0.636	0.645	0.745	0.752	0.760	0.696	0.679	0.595	0.530	
	taud	1.696	1.579	1.603	1.624	1.654	1.509	1.548	1.516	1.592	1.568	1.663	1.744	
	Ebn at Noon	191	191	209	218	219	197	194	189	191	176	176	186	
	Edh at Noon	60	75	79	81	80	92	88	89	79	75	62	54	
All-Sky Solar Radiation	RadAvg	771	921	1288	1551	1686	1650	1470	1359	1132	980	806	738	
	RadStd	90	172	137	136	139	135	159	182	133	141	134	102	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.72	-0.76	-0.94	N/A	N/A	N/A	N/A

Nomenclature: See separate page